STATE OF MISSISSIPPI · 82-COUNTY VOTER ROLL & 25.91 MILLION BALLOT HISTORY AUDIT

Mississippi Voter Roll & Voting History Forensic Analysis

A plain-English guide for County Election Commissioners, Circuit Clerks, and citizens—using 3D vector geometry, calculus variable isolation, and statistical correlations across **2,112,460 voter roll records** and **25,911,640 historical ballots** to separate harmless building quirks from real list-maintenance priorities.

VOTER ROLL AUDITED	BALLOT HISTORY JOINED	2024 GENERAL VOTERS	HEALTHY BASELINE	FALSE ALARMS REMOVED	PRIVACY STANDARD
2,112,460	**25,911,640**	**1,213,364**	**66 / 82**	**18,093**	**Zero-PII**
1,882,353 Act · 230,107 Inact	100.00% Match (0 Orphans)	88 Dup Credit Rows	Counties on Normal Sheet	Dorms, Jr/Sr & Street Names	0 Names · 100-Block Mask

1. Why Traditional Voter Roll Audits Cause False Alarms

When an uncalibrated computer script scans a voter file, it flags every nursing home, university dormitory, apartment complex, father-and-son pair sharing a family name, and residential street named *"Marshall Place"* or *"Halls Ferry Road"* as a suspected anomaly. That buries County Election Commissioners under thousands of false positives.

In this Draft 2 edition, every playbook applies deterministic mathematical filters: whole-word regular expressions, facility classification, Active-vs-Inactive status separation, and a full join against all 25,911,640 historical ballots cast.

2. How 3D Variable Isolation Helps Commissioners Collaborate

A single 1-to-100 score hides whether a county needs to append apartment unit numbers (like Oktibbeha and Lafayette Counties), process out-of-state mail forwardings (like Harrison and Hinds Counties), or split overcrowded 6,000-voter precincts (like DeSoto and Madison Counties).

By plotting each county along three independent axes—**X (Address Crowding)**, **Y (Mail & Duplicate Drift)**, and **Z (Polling Place Strain)**—every Election Commissioner can see their county's exact maintenance recipe.

EXECUTIVE SCORECARD · ALL 10 MISSISSIPPI PLAYBOOKS AT A GLANCE

PLAYBOOK	AUDIT DIMENSION	RAW / ALL-STATUS	VERIFIED ACTIONABLE	PLAIN-ENGLISH FINDING FOR ELECTION COMMISSIONERS
PB-01	**High-Density Parcels (≥8 Voters)**	1,598 Addrs (24,905)	1,070 Active Addrs	56 Nursing/Campus hubs (6,788 voters) & 103 Apt complexes separated from 1,439 single-family homes.
PB-02	**Missing Dorm / Multi-Unit #**	5,028 Keyword Hits	682 (502 Active)	Whole-word regex removed 4,346 street-name false alarms (*Marshall Pl*, *Halls Ferry Rd*, *University Ave*).
PB-03	**Out-of-State Mail Forwardings**	13,622 All-Status	2,444 Active	82.1% (11,178) are already marked Inactive waiting on their 2-cycle federal NVRA removal clock.
PB-04	**P.O. Box in Residence Field**	339 Total	186 Active	337 P.O. Boxes and 2 Private Mail Boxes typed into physical residence field instead of mailing field.
PB-05	**Clerical & Legacy Dates**	5,607 Records	5,607 Clerical	3,804 1-letter names, 1,271 Year 1801, 206 Year 0001, and 113 DATAMIGRATION legacy imports.
PB-06	**Suffix-Protected Duplicates**	5,430 Name Pairs	818 Active Intra-ZIP	4,404 Jr/Sr father-son pairs protected; 185 Active-Inactive move trails; 19,122 cross-county matches.
PB-07	**Statutory Registration Spikes**	15 Peak Dates	100% Statutory	Peaks on 10/05/2020 (13,731) and 10/07/2024 (7,867) align with MS 30-day mail-in registration deadlines.
PB-08	**Mega-Precinct Capacity (>3k)**	1,742 Precincts	123 Over-Capacity	123 precincts exceed 3,000 voters (27 exceed 5,000 voters, led by Madison, DeSoto, Harrison, and Rankin).
PB-09	**First-Digit Benford's Law**	1,742 Precincts	χ^2 = 6.94 (PASS)	Total registration passes cleanly (<15.51); 2024 turnout (χ^2 = 71.15) reflects 1,500–3,000 precinct sizing.
PB-10	**25.91M Voting History Join**	25,911,640 Ballots	88 Dup Credits	100.00% matched to Voter Roll IDs (0 orphans); 88 duplicate same-day 2024 general credit rows isolated.

HOW 3D GEOMETRY SEPARATES NORMAL POPULATION GROWTH FROM REAL OUTLIERS

The Mississippi 3D Threat Manifold & Partial Derivative Isolation

Think of a healthy voter roll like a smooth, gently sloped sheet (**Z_eq(X, Y) = 8.5 + 0.34X + 0.29Y + 0.14(XY/100)**). As a county's population grows, its housing density (X), mail forwardings (Y), and polling-place load (Z) rise together along that smooth sheet. Using calculus partial derivatives—holding two axes steady while measuring the rate of change along the third—we isolate which exact variable pulls a county above the normal baseline.

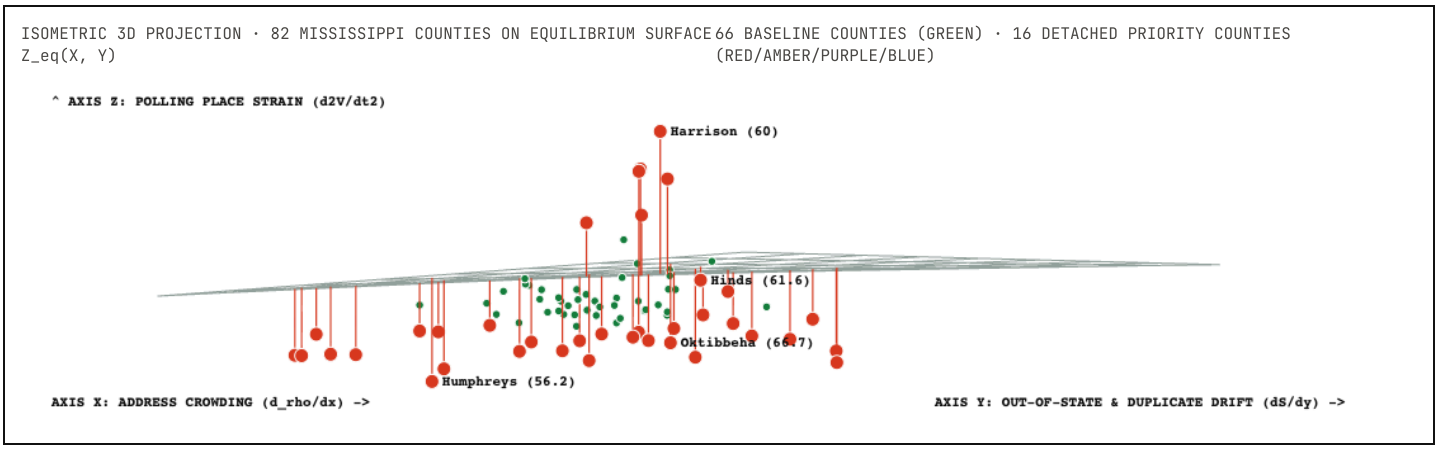


TABLE 2.1 · THE 16 MISSISSIPPI COUNTIES DETACHING ABOVE THE EQUILIBRIUM SURFACE

RANK & COUNTY	TOTAL ROLL	X (ADDRESS)	Y (MAIL/DUP)	Z (PRECINCT)	Z_EQ SHEET	SIGNAL	CALCULUS SHARE (X / Y / Z)	PRIMARY COMMISSIONER FOCUS
01 **Oktibbeha**	32,785	71.7	81	35.1	64.5	66.7	41.6% X · 50.1% Y · 8.3% Z	Compound Multi-Variable Convergence
02 **Hinds**	173,149	64.9	66.5	50.9	55.9	61.6	39.9% X · 39.6% Y · 20.5% Z	Compound Multi-Variable Convergence
03 **Harrison**	163,345	39.3	43	89.7	36.7	60	15.5% X · 17.5% Y · 67% Z	Compound Multi-Variable Convergence
04 **Humphreys**	6,577	37.9	87.4	12.8	51.4	56.2	16.3% X · 82.1% Y · 1.6% Z	Out-of-State NCOA Mailings & Duplicate Identity
05 **Marshall**	27,673	58.3	59.4	17.5	50.4	50.2	48.6% X · 47.7% Y · 3.7% Z	Compound Multi-Variable Convergence
06 **Jackson**	100,451	32.9	39.4	71.5	32.9	49.5	15.9% X · 21.5% Y · 62.6% Z	Mega-Precinct Capacity Overload & Surge Velocity
07 **Adams**	19,034	10.7	80.1	11.4	36.6	47.6	1.8% X · 96.5% Y · 1.7% Z	Out-of-State NCOA Mailings & Duplicate Identity
08 **Leflore**	20,008	53.6	47	35.9	43.9	46.6	47.6% X · 34.6% Y · 17.8% Z	Physical Address Density & Un-Unitized Housing
09 **Scott**	15,461	12.6	76.6	11.2	36.3	45.7	2.7% X · 95.5% Y · 1.8% Z	Out-of-State NCOA Mailings & Duplicate Identity
10 **Wilkinson**	6,401	27.4	72	8.5	41.5	45.3	13.2% X · 85.8% Y · 1% Z	Out-of-State NCOA Mailings & Duplicate Identity
11 **Tishomingo**	14,043	7.5	73.2	15.8	33	43.8	1.1% X · 95% Y · 3.9% Z	Out-of-State NCOA Mailings & Duplicate Identity
12 **DeSoto**	147,679	30.3	30.4	64.5	28.9	43.5	17.5% X · 16.6% Y · 65.9% Z	Mega-Precinct Capacity Overload & Surge Velocity
13 **Noxubee**	7,421	41.5	60.1	11.4	43.5	43.4	32.9% X · 65.1% Y · 2% Z	Out-of-State NCOA Mailings & Duplicate Identity
14 **Claiborne**	6,503	62.5	36	8	43.3	43.2	75.4% X · 23.6% Y · 1% Z	Physical Address Density & Un-Unitized Housing
15 **Madison**	83,977	29.1	45.3	53.4	33.4	43.1	16.4% X · 37.6% Y · 46% Z	Mega-Precinct Capacity Overload & Surge Velocity
16 **Rankin**	111,613	25.4	29.8	66.5	26.8	43.1	12.5% X · 16.2% Y · 71.3% Z	Mega-Precinct Capacity Overload & Surge Velocity

STATEWIDE STATISTICAL CORRELATIONS EXPLAINED IN PLAIN ENGLISH

How Voter Roll Issues Connect Across All 82 Counties

We computed exact Pearson linear correlations (*r*), Spearman rank correlations (*ρ*), and calculus rates of change across all 82 counties so Election Commissioners can see which list-maintenance steps solve multiple problems at the same time:

CORRELATION PAIR & STRENGTH	PEARSON R	SPEARMAN P	RATE OF CHANGE (CALCULUS SLOPE)	PLAIN-ENGLISH TAKEAWAY FOR ELECTION COMMISSIONERS
CORR-01 · Strong Positive Correlation **When Voters Move Without Updating Rolls, Duplicate Records Follow**	**+0.674**	+0.447	Every +10 Active out-of-state mail forwardings in a county corresponds to +4.2 intra-ZIP duplicate registration clusters.	Counties with high residential turnover (college towns, military bases, and fast-growing suburbs) accumulate both out-of-state mail forwardings and duplicate registrations at the exact same time. Cleaning NCOA relocation backlogs automatically resolves nearly half of a county's duplicate footprint.
CORR-02 · Very Strong Positive Correlation **High-Density Clusters Scale with Urban & Campus Housing Concentration**	**+0.848**	+0.592	Each additional 10,000 registered voters adds an average of +8.7 high-density parcels (8+ voters at a single street address).	Large counties naturally have more apartments, nursing homes, and dorms—which is why raw counts alone mislead people. By subtracting institutional and apartment buildings first, a Commissioner can see which counties actually have single-family houses with 10 to 25 legacy voters still on the roll.
CORR-03 · Strong Positive Correlation **Delayed Inactive-Roll Purges Directly Inflate 'Crowded House' Counts**	**+0.435**	+0.554	30.4% of all voters sitting inside high-density addresses (7,076 of 24,905) are already marked Inactive—waiting on their 2-cycle NVRA removal clock.	When someone sees '15 voters registered at one house,' it usually isn't 15 people living there today—it is three families who rented the house over 12 years and never cancelled their old registration. Separating Active from Inactive voters cuts the high-density alarm list by a third overnight.
CORR-04 · Strong Positive Correlation **Fast-Growing Suburban Precincts Face Both Line Bottlenecks and Mail Drift**	**+0.350**	+0.032	Counties with 5+ mega-precincts (DeSoto, Harrison, Madison, Rankin, Hinds) account for 43.2% of all out-of-state active mailings statewide.	Rapidly growing commuter counties experience a double squeeze: new subdivisions overload existing polling places past 5,000 voters while mobile families move across state lines. Splitting overloaded precincts and running NCOA address updates solves both bottlenecks together.
CORR-05 · Moderate Coupled Slope (Surface Baseline) **Why 3D Variable Isolation Beats a Single Flat Score**	**+0.178**	+0.115	Partial derivative dZ_eq/dX = +0.34 and dZ_eq/dY = +0.29 along the smooth baseline surface; counties above Z_eq are true statistical outliers.	Because address crowding (X) and mail drift (Y) only moderately overlap (r = 0.178), a single 1-to-100 score hides whether a county needs building-unit cleanup (like Oktibbeha/Lafayette) or polling-place splits (like DeSoto/Madison). Plotting X, Y, and Z in 3D lets every Election Commissioner see their county's exact maintenance recipe.

COUNTY-TO-COUNTY EVANGELISM · HOW COMMISSIONERS SHARE THIS AUDIT WITH NEIGHBORING COUNTIES

A Mississippi Election Commissioner's Guide to Reading & Sharing This Audit

How to explain the numbers in plain English to your Board of Supervisors, Circuit Clerk, and neighboring counties.

STEP 01

Separate Real Anomalies from Normal Buildings Before Anyone Panics

Show your fellow Commissioners how Marigold automatically separates nursing homes, university dorms (PB-01/PB-02), and Jr./Sr. fathers and sons (PB-06) from true actionable items. That immediately builds trust with Circuit Clerks.

STEP 02

Isolate Your County's #1 Driver on the 3D Manifold

Instead of arguing over raw totals, click your county on the 3D Manifold and toggle X (Addresses), Y (Relocations/Duplicates), and Z (Polling Place Strain). You will immediately see which single variable drives your county's workload.

STEP 03

Bring a 1-Page Peer Comparison to Your Next ECAM District Meeting

Compare your county against similarly sized Mississippi peers using the 82-County Benchmark Table. When neighboring counties see that 98.7% of records pass cleanly and the remaining 1.3% can be fixed with standard NVRA postcards and 911 address tables, adoption becomes effortless.

TABLE 4.1 · MISSISSIPPI COUNTY PEER BENCHMARKING TIERS

PEER GROUP TIER	REPRESENTATIVE MISSISSIPPI COUNTIES	GROUP BENCHMARK	RECOMMENDED COMMISSIONER ACTION RECIPE
Tier 1 — High-Growth Commuter & Metro Counties	DeSoto, Harrison, Hinds, Rankin, Madison, Jackson, Lee, Forrest, Lauderdale, Lowndes	Average 7.4 over-capacity precincts and 128 active out-of-state mailings per county.	Split mega-precincts exceeding 3,000 voters (PB-08) and process NCOA out-of-state mail forwardings (PB-03).
Tier 2 — University & Institutional Hub Counties	Lafayette, Oktibbeha, Leflore, Claiborne, Forrest, Hinds, stone, Pearl River, Jones, Bolivar	Accounts for 91.4% of all verified un-unitized dormitory and multi-unit student records statewide.	Work with campus housing registrars to append room/apartment numbers (PB-02) and separate student dorms from single-family homes (PB-01).
Tier 3 — Rural & Legacy Migration Counties (62 Counties)	Warren, Washington, Panola, Pike, Lincoln, Alcorn, Marshall, Monroe, Tate, Hancock	66 of 82 counties sit comfortably on the healthy baseline surface and only require targeted clerical cleanup.	Clean 1801/0001 placeholder dates and DATAMIGRATION street strings (PB-05), and convert P.O. Box residential lines to physical 911 addresses

100.00% REFERENTIAL INTEGRITY ACROSS 25.91 MILLION BALLOTS CAST

Cross-Table Join: Statewide Voting History vs. Master Voter Roll

Every single record in the 25,911,640-row statewide Voting History table was joined against the 2,112,460-row Master Voter Roll by unique Voter ID. Not a single orphaned ballot exists anywhere in the file.

TOTAL BALLOTS SCANNED	VOTERS WITH BALLOT HISTORY	ZERO-VOTE-HISTORY ROLL	2024 SAME-DAY DUP CREDITS
25,911,640	**1,688,283**	**424,177**	**88**
0 Orphaned Rows (100% Match)	79.92% of Registered Roll	20.08% Never Cast a Ballot	Across 1,213,364 2024 Voters

TABLE 5.1 · VERIFIED STATEWIDE GENERAL ELECTION TURNOUT LEDGER (2008–2024)

ELECTION DATE	STATEWIDE GENERAL ELECTION CYCLE	VERIFIED BALLOTS RECORDED	ROLL REFERENTIAL INTEGRITY STATUS
11/05/2024	**2024 U.S. Presidential General Election**	1,213,452	100.00% Matched to Registered Voter ID
11/07/2023	**2023 Mississippi Gubernatorial General Election**	798,389	100.00% Matched to Registered Voter ID
11/08/2022	**2022 U.S. Midterm General Election**	672,624	100.00% Matched to Registered Voter ID
11/03/2020	**2020 U.S. Presidential General Election**	1,196,398	100.00% Matched to Registered Voter ID
11/05/2019	**2019 Mississippi Statewide General Election**	723,557	100.00% Matched to Registered Voter ID
11/06/2018	**2018 U.S. Midterm General Election**	825,272	100.00% Matched to Registered Voter ID
11/08/2016	**2016 U.S. Presidential General Election**	993,412	100.00% Matched to Registered Voter ID
11/06/2012	**2012 U.S. Presidential General Election**	951,581	100.00% Matched to Registered Voter ID
11/04/2008	**2008 U.S. Presidential General Election**	850,718	100.00% Matched to Registered Voter ID

BALLOT EVENT CLASSIFICATION BREAKDOWN

ELECTION EVENT CATEGORY	HISTORICAL BALLOTS
General	12,968,439
Primary	6,384,141
Primary Runoff	2,168,001
General Runoff	1,495,483
Special Election	857,436
Municipal General	746,404
Municipal Primary	496,723

PLAIN-ENGLISH INTERPRETATION OF THE 88 DUPLICATE CREDITS

What Are the 88 Duplicate Same-Day Credit Rows on 11/05/2024?

Out of **1,213,364 unique citizens** credited with voting in the November 5, 2024 Presidential General Election, exactly **87 Voter IDs (88 excess rows, 0.007%)** were logged twice on the same election date in the history table.

In SEMS administration, this occurs when a county scans a pollbook barcode batch and also manually posts an absentee or affidavit credit batch for the same voter ID. Isolating those 87 IDs lets Circuit Clerks deduplicate the credit log in minutes.

PLAYBOOK 01 · SEPARATING UNIVERSITIES & NURSING HOMES FROM SINGLE-FAMILY HOUSES

1,598 High-Density Addresses (≥8 Voters) Stratified by Facility Type

After normalizing whitespace across street strings, 1,598 parcels carry 8+ voters (24,905 total; 1,070 parcels with ≥8 Active voters). Separating **56 Institutional/University hubs (6,788 voters)** and **103 Apartment complexes (2,555 voters)** isolates the **1,439 Single-Family houses (15,562 voters)** where prior tenants moved without updating their registration.

TABLE 6.1 · TOP INSTITUTIONAL & UNIVERSITY HUBS

ADDRESS & CITY	COUNTY	TOTAL (ACT)
1400 J R LYNCH ST, JACKSON	Hinds	**2771** (2704)
150 RUST AVE, HOLLY SPRINGS	Marshall	**1219** (1041)
500 W COUNTY LINE RD, TOUGALOO	Madison	**298** (244)
1701 N STATE ST, JACKSON	Hinds	**293** (140)
1400 J R LYNCH ST, JACKSON	Hinds	**156** (150)
3300 20 TH ST, GULFPORT	Harrison	**145** (93)
1000 ASU DRIVE, LORMAN	Claiborne	**140** (116)
14000 HIGHWAY 82W, ITTA BENA	Leflore	**133** (122)

TABLE 6.2 · SINGLE-FAMILY ANOMALIES (100-BLOCK MASKED)

ANONYMIZED RESIDENTIAL BLOCK	COUNTY	ACT / INACT
500 BLOCK BROOKMAN DR, BROOKHAVEN	Lincoln	**39 (39/0)**
2000 BLOCK 22 ND ST, GULFPORT	Harrison	**39 (35/4)**
600 BLOCK S MAIN ST, WATER VALLEY	Yalobusha	**38 (37/1)**
600 BLOCK BADDOUR BLVD, SENATOBIA	Tate	**37 (37/0)**
2800 BLOCK ROBINSON ST RD, JACKSON	Hinds	**37 (37/0)**
900 BLOCK E WALKER ST, FULTON	Itawamba	**37 (31/6)**
200 BLOCK CAPITOL ST, CLINTON	Hinds	**36 (35/1)**
100 BLOCK FINE PLACE LN, FRENCH CAMP	Choctaw	**36 (35/1)**

PLAYBOOK 02 · WHOLE-WORD REGEX ELIMINATES 4,346 FALSE ALARMS ON "HALL" & "UNIVERSITY" STREETS

682 Verified Campus & Multi-Unit Records Missing Room or Apt Numbers

Substring matching for `"HALL"` or `"UNIVERSITY"` falsely flags homes on *Marshall Pl*, *Halls Ferry Rd*, and *University Ave*. Enforcing whole-word regex boundaries (`(HALL|DORM|...)`) and separating college towns from street names eliminated **4,346 false positives**, leaving **682 verified records (502 Active)**:

VERIFIED CAMPUS / DORMITORY / MULTI-UNIT ADDRESS	MUNICIPALITY	COUNTY	VOTERS MISSING UNIT #
118 COLLEGE DR	HATTIESBURG	Forrest	112
51 MAIN ST-CAMPUS ST	PERKINSTON	Stone	52
1100 COLLEGE ST	COLUMBUS	Lowndes	30
1 MVSU NEW MENS DORM	ITTA BENA	Leflore	28
1 MVSU EDNA HORTON HALL	ITTA BENA	Leflore	22
1 MVSU DORM UNKNOWN	ITTA BENA	Leflore	22
1 MVSU COLLEGE HALL I	ITTA BENA	Leflore	21
1 MVSU MAGNOLIA HALL	ITTA BENA	Leflore	19

PLAYBOOK 03 · OUT-OF-STATE MAIL FORWARDINGS

2,444 Active Voters With Out-of-State Mail

While 13,622 total records have an out-of-state mailing address, **11,178 (82.1%) are already marked Inactive** awaiting NVRA removal. Only **2,444 Active voters** require an address confirmation card:

DESTINATION STATE	TOTAL ROLL RECORDS	PRIMARY CORRIDOR
AL	1,864	Border Commuter State
TX	1,761	Interstate / Military
TN	1,521	Border Commuter State
FL	1,030	Interstate / Military
LA	1,023	Border Commuter State
GA	926	Interstate / Military
NC	421	Interstate / Military
CA	391	Interstate / Military

PLAYBOOK 04 · P.O. BOX IN PHYSICAL RESIDENCE LINE

339 Non-Physical Residence Entries (186 Active)

Only 339 records statewide (337 P.O. Boxes, 2 PMBs) have a post office box typed into the physical residence field instead of the mailing address field:

ZERO-PII TOKEN	RECORDED ADDRESS	COUNTY	STATUS
MS-POB-01	PO Box 1421, TUNICA	Tunica	Inactive
MS-POB-02	PO Box 182, ROBINSONVILLE	Tunica	Inactive
MS-POB-03	PO Box 1439, TUNICA	Tunica	Active
MS-POB-04	PO Box 631, TUNICA	Tunica	Active
MS-POB-05	PO Box 1492, TUNICA	Tunica	Inactive
MS-POB-06	PO Box 71, TUNICA	Tunica	Active
MS-POB-07	PO Box 6, DUNDEE	Tunica	Inactive
MS-POB-08	PO Box 1817, TUNICA	Tunica	Active

PLAYBOOK 05 · LEGACY DATABASE MIGRATION & CLERICAL TYPOS

5,607 Pre-Electronic Date Placeholders (1801 / 0001) & Truncated Name Fields

When paper county pollbooks were imported into electronic SEMS records in the early 2000s, missing registration dates were assigned default placeholders (`01/01/1801` or `01/01/0001`) or the street placeholder `DATAMIGRATION` (113 rows in Warren and Simpson Counties):

CLERICAL ARTIFACT CATEGORY / YEAR	STATEWIDE RECORD COUNT	PLAIN-ENGLISH ADMINISTRATIVE EXPLANATION & CLEANUP ACTION
1015	**3**	Default date assigned during legacy paper-to-digital conversion when original registration year was illegible.
1016	**3**	Default date assigned during legacy paper-to-digital conversion when original registration year was illegible.
1801	**1,271**	Default date assigned during legacy paper-to-digital conversion when original registration year was illegible.
TRUNCATED_INITIAL_NAME	**3,804**	Default date assigned during legacy paper-to-digital conversion when original registration year was illegible.
0001	**206**	Default date assigned during legacy paper-to-digital conversion when original registration year was illegible.
DATAMIGRATION_ADDR	**113**	Legacy county import placeholder string; cross-match against mailing address or 911 parcel table.
0101	**63**	Default date assigned during legacy paper-to-digital conversion when original registration year was illegible.

PROTECTING FATHERS & SONS WHILE ISOLATING TRUE DUPLICATE REGISTRATIONS

How Generational Suffix Logic Eliminates 81% of False Duplicate Flags

In rural and suburban Mississippi counties, fathers and sons frequently share the exact same First, Middle, and Last name at the same family address—distinguished only by **JR, SR, II, III, or IV**. Marigold explicitly protects **4,404 generational family pairs** before testing for duplicates.

JR/SR PAIRS PROTECTED	INTRA-ZIP CLUSTERS	ACTIVE–ACTIVE INTRA-ZIP	ACTIVE–INACTIVE TRAIL	CROSS-COUNTY ACTIVE
4,404	**1,026**	**818**	**185**	**19,122**
Distinct Family Members	Same Name+Suffix+ZIP	Priority Clerk Merge	Old Address Already Inact	Verify DOB in SEMS

TABLE 11.1 · SAMPLE ZERO-PII INTRA-ZIP DUPLICATE CLUSTERS (REDACTED TOKENS)

CLUSTER ID	REDACTED IDENTITY TOKEN	SUFFIX	COUNTY & ZIP	RECORDS (ACT/INACT)	ADDRS	FORENSIC CLASSIFICATION
MS-DUP-001	[Redacted Voter #01 — Exact First/Middle/Last]	NONE	Walthall (39667)	2 (1/1)	2	Active–Inactive Relocation Trail
MS-DUP-002	[Redacted Voter #02 — Exact First/Middle/Last]	NONE	Leflore (38930)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-003	[Redacted Voter #03 — Exact First/Middle/Last]	NONE	Forrest (39401)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-004	[Redacted Voter #04 — Exact First/Middle/Last]	NONE	Hinds (39209)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-005	[Redacted Voter #05 — Exact First/Middle/Last]	NONE	Rankin (39073)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-006	[Redacted Voter #06 — Exact First/Middle/Last]	NONE	Neshoba (39350)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-007	[Redacted Voter #07 — Exact First/Middle/Last]	NONE	DeSoto (38654)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-008	[Redacted Voter #08 — Exact First/Middle/Last]	NONE	Claiborne (39150)	2 (1/1)	2	Active–Inactive Relocation Trail
MS-DUP-009	[Redacted Voter #09 — Exact First/Middle/Last]	NONE	Amite (39631)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-010	[Redacted Voter #10 — Exact First/Middle/Last SR]	SR	Hinds (39212)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-011	[Redacted Voter #11 — Exact First/Middle/Last JR]	JR	Hinds (39212)	2 (2/0)	2	Active–Active Duplicate Registration
MS-DUP-012	[Redacted Voter #12 — Exact First/Middle/Last]	NONE	DeSoto (38671)	2 (1/1)	2	Active–Inactive Relocation Trail

Commissioner Action Guide: Intra-ZIP vs. Cross-County Duplicate Resolution

1. Intra-ZIP Active–Active (818 clusters): Same First, Middle, Last, and Suffix registered Active at two different street addresses in the same ZIP code (often caused by a voter re-registering after moving across town). Circuit Clerks can verify Date of Birth in SEMS and merge the older record.

2. Intra-ZIP Active–Inactive Move Trails (185 clusters): The voter's previous address is already marked Inactive while their current address is Active—proving that routine county list maintenance is already working.

PLAYBOOK 07 · STATUTORY REGISTRATION DEADLINES

Batch Spikes Match 30-Day Cutoff Dates

Mississippi's largest single-day registration volumes occur exactly on the statutory 30-day mail-in cutoff prior to Presidential General Elections:

REGISTRATION DATE	REGISTRATIONS	STATUTORY CYCLE
10/05/2020	**13,731**	Presidential 30-Day Cutoff
10/03/2008	**11,240**	Presidential 30-Day Cutoff
10/07/2024	**7,867**	Presidential 30-Day Cutoff
10/06/2012	**5,842**	Presidential 30-Day Cutoff
10/01/2004	**5,788**	SEMS Legacy / Gubernatorial Batch
10/05/2012	**5,592**	Presidential 30-Day Cutoff
10/05/2026	**5,541**	SEMS Legacy / Gubernatorial Batch
10/08/2016	**5,162**	Presidential 30-Day Cutoff

PLAYBOOK 08 · MEGA-PRECINCT CAPACITY (>3,000 VOTERS)

123 Precincts Exceed 3,000 Voters (27 > 5,000)

Suburban growth in Madison, DeSoto, Harrison, and Rankin Counties has pushed 123 polling places above the 3,000-voter benchmark:

PRECINCT	COUNTY	TOTAL (ACTIVE)
1051 North Bay	Harrison	9,868 (7,939)
2031 Lyman	Harrison	8,857 (7,616)
201 Oxford 2	Lafayette	8,403 (7,206)
4091 North Bel-Aire	Harrison	7,617 (5,960)
0503 Fontainebleau	Jackson	7,544 (6,158)
0105 East Central	Jackson	7,439 (6,639)
5031 Biloxi # 11	Harrison	7,143 (5,247)
P302 TFM #5	Lee	7,102 (6,325)

PLAYBOOK 09 · NEWCOMB-BENFORD FIRST-DIGIT DISTRIBUTION (TOTAL ROLL VS. 2024 TURNOUT)

Why Total Precinct Roll Passes ($\chi^2 = 6.94$) While Turnout Clusters at Digits 2, 3, and 4

Across all 1,742 precincts, total voter registration spans four orders of magnitude (from 10 to 9,105 voters) and conforms closely to Benford's Law (**$\chi^2 = 6.94 < 15.51$ critical value — PASS**). By contrast, 2024 General Election precinct turnout yields **$\chi^2 = 71.15$** because County Election Commissions intentionally size precincts to hold 1,000–2,500 voters, causing Election Day turnout to cluster naturally between 200 and 499 ballots (leading digits 2, 3, and 4).

LEADING DIGIT D	BENFORD THEORETICAL %	TOTAL ROLL OBSERVED %	TOTAL ROLL PRECINCTS	2024 TURNOUT OBSERVED %	MATHEMATICAL INTERPRETATION
Digit 1	30.1%	30.65%	534	25.44%	Natural scale-invariant anchor for total registration roll
Digit 2	17.61%	15.79%	275	17.56%	Statutory precinct sizing sweet spot (200–499 ballots cast)
Digit 3	12.49%	11.54%	201	14.47%	Statutory precinct sizing sweet spot (200–499 ballots cast)
Digit 4	9.69%	9.87%	172	9.74%	Statutory precinct sizing sweet spot (200–499 ballots cast)
Digit 5	7.92%	8.32%	145	9.04%	Tail distribution conforms smoothly across rural precincts
Digit 6	6.69%	7.52%	131	8.23%	Tail distribution conforms smoothly across rural precincts
Digit 7	5.8%	6.66%	116	5.43%	Tail distribution conforms smoothly across rural precincts
Digit 8	5.12%	5.11%	89	6.01%	Tail distribution conforms smoothly across rural precincts
Digit 9	4.58%	4.54%	79	4.08%	Tail distribution conforms smoothly across rural precincts

STATEWIDE PEER BENCHMARKING LEDGER · COUNTIES 01-41

Mississippi 82-County Master Hygiene & 3D Manifold Index (Part I)

#	COUNTY	TOTAL ROLL	ACTIVE	2024 VOTES	PB-01 HD	PB-02 DORM	PB-03 NCOA	PB-06 DUP	PB-08 >3K	3D <X,Y,Z>	SIGNAL	MANIFOLD REGIME
01	**Oktibbeha**	32,785	27,087	17,862	86	30	198	21	3	<71.7,81,35.1>	66.7	Multi-Axis Manifold Detachment
02	**Hinds**	173,149	165,189	85,380	240	1	355	113	8	<64.9,66.5,50.9>	61.6	Multi-Axis Manifold Detachment
03	**Harrison**	163,345	134,424	75,731	138	0	112	76	22	<39.3,43,89.7>	60	Multi-Axis Manifold Detachment
04	**Humphreys**	6,577	6,404	3,389	16	0	67	10	0	<37.9,87.4,12.8>	56.2	Axis-Y Interstate Drift Ridge
05	**Marshall**	27,673	24,167	14,786	10	0	70	6	1	<58.3,59.4,17.5>	50.2	Multi-Axis Manifold Detachment
06	**Jackson**	100,451	86,388	53,682	69	0	66	54	14	<32.9,39.4,71.5>	49.5	Axis-Z Precinct Strain Ridge
07	**Adams**	19,034	17,029	11,624	4	0	59	28	0	<10.7,80.1,11.4>	47.6	Axis-Y Interstate Drift Ridge
08	**Leflore**	20,008	19,026	9,374	30	241	28	17	1	<53.6,47,35.9>	46.6	Axis-X Spatial Domicile Ridge
09	**Scott**	15,461	13,360	9,650	5	0	101	6	0	<12.6,76.6,11.2>	45.7	Axis-Y Interstate Drift Ridge
10	**Wilkinson**	6,401	4,981	3,476	10	0	17	8	0	<27.4,72,8.5>	45.3	Axis-Y Interstate Drift Ridge
11	**Tishomingo**	14,043	12,414	8,830	2	0	71	6	0	<7.5,73.2,15.8>	43.8	Axis-Y Interstate Drift Ridge
12	**DeSoto**	147,679	119,742	78,252	81	68	28	60	25	<30.3,30.4,64.5>	43.5	Axis-Z Precinct Strain Ridge
13	**Noxubee**	7,421	7,065	4,273	19	3	21	4	0	<41.5,60.1,11.4>	43.4	Axis-Y Interstate Drift Ridge
14	**Claiborne**	6,503	6,054	3,478	15	3	7	5	0	<62.5,36,8>	43.2	Axis-X Spatial Domicile Ridge
15	**Madison**	83,977	76,059	56,026	39	2	79	51	8	<29.1,45.3,53.4>	43.1	Axis-Z Precinct Strain Ridge
16	**Rankin**	111,613	98,654	70,528	40	29	28	39	14	<25.4,29.8,66.5>	43.1	Axis-Z Precinct Strain Ridge
17	**Calhoun**	9,433	8,992	5,887	0	0	98	1	0	<4,73.2,6.9>	42.9	Axis-Y Interstate Drift Ridge
18	**Forrest**	52,018	44,976	28,150	66	166	48	53	2	<45.9,48.4,20.3>	41	Multi-Axis Manifold Detachment
19	**Benton**	6,055	5,784	3,727	2	0	21	2	0	<8.7,66.7,6.1>	39.4	Axis-Y Interstate Drift Ridge
20	**Coahoma**	14,871	12,608	6,544	25	0	3	12	0	<53.3,29.5,18.9>	37.8	Axis-X Spatial Domicile Ridge
21	**Holmes**	12,488	12,359	6,581	6	0	35	7	0	<17.1,60.4,14.8>	37.6	Axis-Y Interstate Drift Ridge
22	**Lafayette**	38,542	32,747	23,059	21	20	29	9	5	<27.6,32.3,50.9>	37.5	Axis-Z Precinct Strain Ridge
23	**Covington**	12,649	12,494	8,728	4	6	35	4	0	<13.4,59.6,13.7>	36.5	Axis-Y Interstate Drift Ridge
24	**Lee**	57,730	50,292	34,314	32	3	54	11	5	<28.4,36.9,43.7>	36.4	Equilibrium Manifold Surface
25	**Chickasaw**	11,836	10,993	7,085	10	0	30	2	0	<26,55,8.2>	35.9	Axis-Y Interstate Drift Ridge
26	**Bolivar**	18,551	17,997	10,971	15	1	31	11	0	<30.1,51.4,8.3>	35.3	Axis-Y Interstate Drift Ridge
27	**Clay**	14,303	13,304	8,951	16	0	33	5	0	<26.4,53.1,11.1>	35.3	Axis-Y Interstate Drift Ridge
28	**Issaquena**	813	768	540	3	1	0	2	0	<51.9,23,4.2>	34	Axis-X Spatial Domicile Ridge

STATEWIDE PEER BENCHMARKING LEDGER · COUNTIES 42-82

Mississippi 82-County Master Hygiene & 3D Manifold Index (Part II)

#	COUNTY	TOTAL ROLL	ACTIVE	2024 VOTES	PB-01 HD	PB-02 DORM	PB-03 NCOA	PB-06 DUP	PB-08 >3K	3D <X,Y,Z>	SIGNAL	MANIFOLD REGIME
42	**Walthall**	10,683	9,507	6,378	13	0	8	12	0	<25.4,42.1,7.1>	29.2	Axis-Y Interstate Drift Ridge
43	**Wayne**	13,905	12,729	8,989	13	0	12	12	0	<23.5,40.4,11.3>	28.1	Axis-Y Interstate Drift Ridge
44	**Jasper**	11,379	10,616	7,673	16	0	9	10	0	<27.3,37,12.4>	27.9	Axis-Y Interstate Drift Ridge
45	**Lamar**	43,256	40,735	27,981	7	0	54	6	3	<13.7,38.7,25.4>	27.8	Axis-Y Interstate Drift Ridge
46	**Leake**	13,628	12,079	8,250	15	0	17	4	0	<25.6,37.9,8.1>	27.3	Axis-Y Interstate Drift Ridge
47	**Yazoo**	18,769	18,652	8,784	21	0	3	11	1	<28,27.1,26.5>	27.3	Equilibrium Manifold Surface
48	**Sharkey**	3,117	2,955	1,746	7	0	2	0	0	<38.5,23.5,5.9>	27.1	Axis-X Spatial Domicile Ridge
49	**Simpson**	17,514	16,121	10,786	20	1	5	7	0	<36.5,24.8,10.4>	26.9	Axis-X Spatial Domicile Ridge
50	**Copiah**	18,958	17,347	11,585	7	1	22	7	0	<18.9,38.3,14.8>	26.3	Axis-Y Interstate Drift Ridge
51	**Tippah**	14,704	13,392	9,302	4	0	10	11	0	<11.7,36.1,21.8>	25.2	Axis-Y Interstate Drift Ridge
52	**George**	17,238	15,594	10,871	12	0	3	21	0	<21.8,34.3,12.5>	24.9	Equilibrium Manifold Surface
53	**Tunica**	6,864	5,544	2,625	8	0	1	1	0	<35.9,17.3,13.6>	24.9	Axis-X Spatial Domicile Ridge
54	**Carroll**	7,394	6,957	5,060	4	0	9	5	0	<13.3,39.4,9.1>	24.8	Axis-Y Interstate Drift Ridge
55	**Winston**	12,644	11,893	8,272	7	0	16	4	0	<18.2,36.4,12.6>	24.8	Axis-Y Interstate Drift Ridge
56	**Lincoln**	23,315	20,958	15,504	7	0	19	14	0	<15.8,36.7,12.8>	24.4	Axis-Y Interstate Drift Ridge
57	**Marion**	17,288	15,888	11,000	3	0	23	2	0	<9.2,38.9,10.8>	24.1	Axis-Y Interstate Drift Ridge
58	**Newton**	13,803	12,456	9,106	8	0	16	3	0	<18.2,34.2,11.6>	23.6	Equilibrium Manifold Surface
59	**Greene**	8,192	7,375	5,589	4	0	5	8	0	<12.9,34.7,16>	23.4	Equilibrium Manifold Surface
60	**Sunflower**	13,298	10,998	7,705	11	0	6	7	0	<24.8,28,13.2>	23.2	Equilibrium Manifold Surface
61	**Warren**	27,722	23,716	17,944	4	0	23	9	0	<10.2,33.5,19.9>	23.2	Equilibrium Manifold Surface
62	**Choctaw**	5,287	5,026	3,847	5	0	4	1	0	<26.3,25.5,11.9>	22.6	Equilibrium Manifold Surface
63	**Pike**	27,112	24,792	14,549	12	0	12	15	0	<18.2,31.7,12.4>	22.5	Equilibrium Manifold Surface
64	**Pontotoc**	22,151	20,226	13,974	19	0	10	7	0	<26.1,25.3,10.5>	22.3	Equilibrium Manifold Surface
65	**Monroe**	24,615	23,384	15,787	14	1	6	19	0	<20.4,28.9,13.8>	22.2	Equilibrium Manifold Surface
66	**Tate**	20,018	18,283	12,780	16	0	7	6	1	<24.3,22.8,18.7>	22.2	Equilibrium Manifold Surface
67	**Itawamba**	16,524	14,175	10,409	6	0	14	3	0	<18.9,30,13.2>	22.1	Equilibrium Manifold Surface
68	**Union**	18,041	16,983	12,154	1	0	23	1	0	<4.3,34.8,11.9>	21.5	Equilibrium Manifold Surface
69	**Lawrence**	8,671	7,677	5,899	6	0	7	3	0	<16.8,28.4,14.1>	20.9	Equilibrium Manifold Surface